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REAL-TIME TREMOR ANALYSIS (robust demo) Parkinson's Tremor Monitoring System - MATLAB

```
clc; clear; close all;
```

----- USER DATA INPUT -----

If you already have variables named `mpu` and `flex` in workspace, the script will use them. Otherwise it uses small synthetic demo signals.

```
if exist('mpu','var') && exist('flex','var')
    mpu_data  = mpu(:);   % force column
    flex_data = flex(:);
    fprintf("Using workspace variables 'mpu' and 'flex'.\n");
else
    % Demo synthetic signals (replace with your data later)
    fs = 50;                    % sampling rate
    t  = (0:1/fs:10-1/fs)';     % 10 seconds
    % synthetic tremor ~5 Hz for demo
    mpu_data  = 0.5*sin(2*pi*5*t) + 0.05*randn(size(t));
    flex_data = 0.3*sin(2*pi*5*t + pi/6) + 0.05*randn(size(t));
    fprintf("No 'mpu'/'flex' in workspace — running demo signals (10 s, 50
Hz).\n");
end

% If fs was not set above, set default
if ~exist('fs','var'), fs = 500; end
```

No 'mpu'/'flex' in workspace — running demo signals (10 s, 50 Hz).

1. Pre-check lengths

```
N1 = length(mpu_data);
N2 = length(flex_data);
if N1 ~= N2
    warning("mpu and flex data lengths differ: %d vs %d. Truncating to min
length.", N1, N2);
    Nmin = min(N1,N2);
    mpu_data  = mpu_data(1:Nmin);
    flex_data = flex_data(1:Nmin);
end
N = length(mpu_data);
```

2. Band-pass filter (3–7 Hz)

If `bandpass` function exists use it; otherwise implement a simple FIR bandpass.

```
if exist('bandpass','file')
    filtered_mpu  = bandpass(mpu_data,  [3 7], fs);
    filtered_flex = bandpass(flex_data, [3 7], fs);
else
    % Simple FIR bandpass using designfilt if available, else butter
    if exist('butter','file')
        [b,a] = butter(4, [3 7]/(fs/2), 'bandpass');   % 4th order Butterworth
        filtered_mpu  = filtfilt(b,a, mpu_data);
        filtered_flex = filtfilt(b,a, flex_data);
    else
        error("No bandpass/butter available. Install Signal Processing
Toolbox or provide filtered data.");
    end
end
```

3. RMS Calculation (manual to avoid toolbox dependency)

```
rms_mpu  = sqrt(mean(filtered_mpu.^2));
rms_flex = sqrt(mean(filtered_flex.^2));
```

4. FFT Calculation (use positive frequencies only)

```
FFT_mpu_full  = fft(filtered_mpu);
FFT_flex_full = fft(filtered_flex);
f_axis = (0:N-1)*(fs/N);            % full axis
% Single-sided spectrum
halfN = floor(N/2)+1;
FFT_mpu  = abs(FFT_mpu_full(1:halfN))/N * 2;
```

```
FFT_flex = abs(FFT_flex_full(1:halfN))/N * 2;
f_single = f_axis(1:halfN);
```

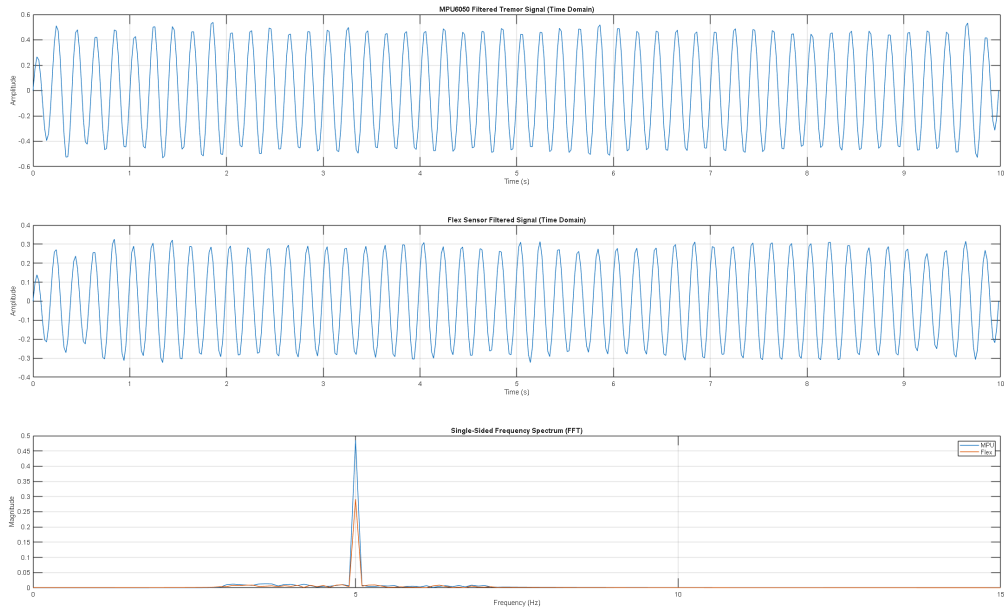
5. Tremor Detection Threshold and Decision Logic

```
rms_threshold =0.25;   % adjust later
% If you want dynamic threshold, you can compute baseline RMS here
mpu_tremor  = rms_mpu  > rms_threshold;
flex_tremor = rms_flex > rms_threshold;

% Combine rules:
% - If either sensor detects tremor -> tremor detected (covers wrist-only or
pill-only)
% - If both detect -> high confidence
if mpu_tremor && flex_tremor
    tremor_status = "Tremor Detected (High confidence)";
elseif mpu_tremor || flex_tremor
    tremor_status = "Tremor Detected";
else
    tremor_status = "No Tremor";
end

fprintf("Tremor Status: %s\n", tremor_status);
fprintf("MPU RMS: %.4f    Flex RMS: %.4f    Threshold: %.4f\n", rms_mpu,
rms_flex, rms_threshold);
```

Tremor Status: Tremor Detected
MPU RMS: 0.3452   Flex RMS: 0.2078   Threshold: 0.2500

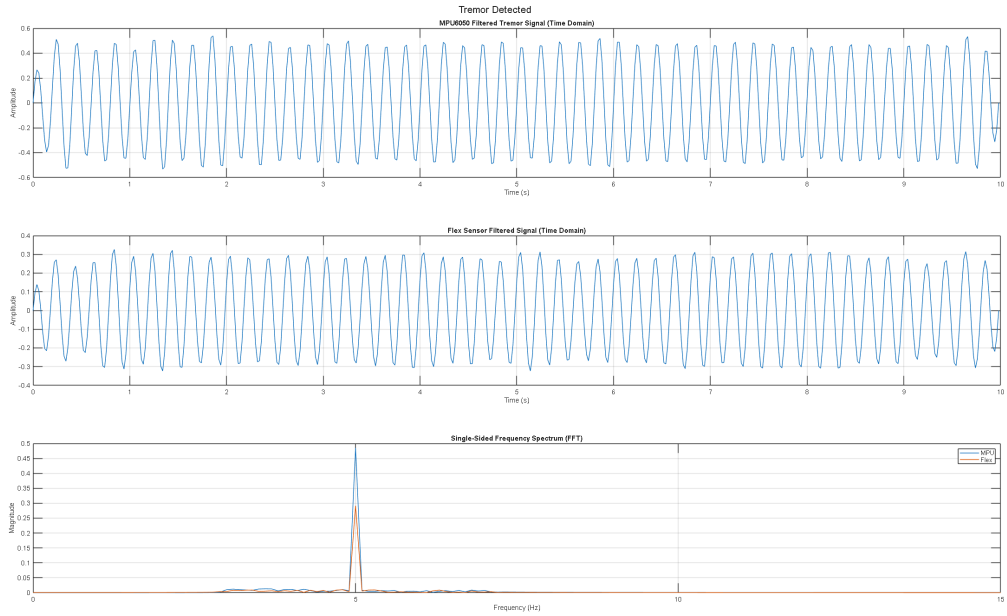
6. Plot Time-Domain and Frequency-Domain

```
figure('Units','normalized','Position',[0.1 0.1 0.7 0.7]);
subplot(3,1,1);
plot((0:N-1)/fs, filtered_mpu); grid on;
title("MPU6050 Filtered Tremor Signal (Time Domain)");
xlabel("Time (s)"); ylabel("Amplitude");

subplot(3,1,2);
plot((0:N-1)/fs, filtered_flex); grid on;
title("Flex Sensor Filtered Signal (Time Domain)");
xlabel("Time (s)"); ylabel("Amplitude");

subplot(3,1,3);
plot(f_single, FFT_mpu); hold on;
plot(f_single, FFT_flex); grid on;
xlim([0 15]);    % PD tremor < 10 Hz, but show up to 15 Hz
legend("MPU","Flex");
title("Single-Sided Frequency Spectrum (FFT)");
xlabel("Frequency (Hz)"); ylabel("Magnitude");
```



7. Mark detection on figure (optional)

```
sgtitle(string(tremor_status));
```



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